

Exercise 3

```
a = 0; % lower limit
H = 30; % upper limit
n = 100; % number of segments
f1 = @(z)force_on_sailboat(z,H);
f2 = @(z)numerator(z,H);
F_integral = integral(f1,a,H);
d_integral = integral(f2,a,H)/integral(f1,a,H);
F_trapezoidal = trap(f1,a,H,n);
d_trapezoidal = trap(f2,a,H,n)/trap(f1,a,H,n);
F_simp = simp(f1,a,H,n);
d_simp = simp(f2,a,H,n)/simp(f1,a,H,n);
fprintf( ...
"Total Force F using integral = %4.4f \n" + ...
"Total Force F using trapezoidal = %4.4f \n" + ...
"Total Force F using simpson = %4.4f \n",F_integral,F_trapezoidal,F_simp)
```

```
Total Force F using integral = 970.4526
Total Force F using trapezoidal = 970.2316
Total Force F using simpson = 970.4523
```

```
fprintf("Line of Action d using integral = %4.4f \n" + ...
"Line of Action d using trapezoidal = %4.4f \n" + ...
"Line of Action d using simpson = %4.4f \n",d_integral,d_trapezoidal, d_simp)
```

```
Line of Action d using integral = 12.1164
Line of Action d using trapezoidal = 12.1191
Line of Action d using simpson = 12.1164
```

Comparing results of simpson and trapezoidal with built-in integral function for F:

```
perect_error_trapezoidal = 100*abs(F_trapezoidal-F_integral)/F_integral;
percent_error_simp = 100*abs(F_simp-F_integral)/F_integral;
fprintf("Percent error for trapezoidal = %f \n" + ...
"Percent error for simpson = %f \n" + ...
"We can see that the simpson 1/3 approximation, when compared with the trapezoidal function
"is closer to the built-in integral function result when calculating F.",perect_error_trapezoidal,percent_error_simp)
```

```
Percent error for trapezoidal = 0.022779
Percent error for simpson = 0.000030
We can see that the simpson 1/3 approximation, when compared with the trapezoidal function,
is closer to the built-in integral function result when calculating F.
```

Calculatin percent error for Line of Action d:

```
p_error_simp = 100*abs(d_integral-d_simp)/d_integral;
p_error_trap = 100*abs(d_integral-d_trapezoidal)/d_integral;
fprintf("Perecent error for Line of Action d trapezoidal = %f \n" + ...
"Percent of error for Line of Action d simpson = %f \n" + ...
"We can see that the approximation to the Line of action d given\n" + ...
"by the simpson function is closer to the integral function result\n" + ...
"with a percent error of %f",p_error_trap,p_error_simp,p_error_simp)
```

```
Perecent error for Line of Action d trapezoidal = 0.021729
```

Percent of error for Line of Action d simpson = 0.000045

We can see that the approximation to the Line of action d given by the simpson function is closer to the integral function result with a percent error of 0.000045

```
function [y] = force_on_sailboat(z,H)
y = 200*(z./(7+z)).*exp(-2.7*z./H);
end

function [y] = numerator(z,H)
y = z.*(200*(z./(7+z)).*exp(-2.7*z./H));
end

function [I] = trap(f,a,b,n)
I=0; % integral initial value
h = (b-a)/n; % step size
for i=1:n
    x = a + i*h;% going from a to b w. step size h
    if (i==0 || i==n)
        I = I + f(x); % evaluate end points
    else
        I = I + 2*f(x);
    end
end
I = I*h/2;
end

function [I] = simp(f,a,b,n)
if (mod(n,2)==0)
    I = 0;
    h = (b-a)/n;
    for i=1:n
        x = a + i*h;
        if (i==0 || i==n)
            I = I + f(x); % endpoints
        elseif (mod(i,2)==0)
            I = I + 2*f(x);
        else
            I = I + 4*f(x);
        end
    end
    I = I*h/3;
else
    disp("number of segments n must be even")
end
end
```